💡 **Useful Formulas for kinematics**

(1) $v = v_0 + at$

(2) $\Delta x = v_0 t + \frac{1}{2}at^2$

(3) $v^2 = {v_0}^2 + 2a\Delta x$

Q1

A $20\,\mathrm{kg}$ table sits on a flat floor. The static friction coefficient is $\mu_s = 0.6$ and the kinetic friction coefficient is $\mu_k = 0.4$.

(a) Calculate the normal force acting on the table.

(b) You push the table with $100\,\mathrm{N}$ of force. Does the table move? What is the friction force pushing back against you right now?

Q2

Based on Q1, now you push harder with $150\,\mathrm{N}$ of force.

(a) Which friction (static or kinetic) acts on it now?

(b) Focus on the X-direction. Calculate the net force.

(c) Calculate the acceleration of the table.

Q3

Two dogs are pulling a toy bone. Dog A pulls West with $60\,\mathrm{N}$. Dog B pulls South with $80\,\mathrm{N}$.

(a) Draw a dot representing the toy bone, and draw these two force vectors acting on it.

(b) Use the Pythagorean theorem to calculate the magnitude of the net force on the toy.

Q4

You are pulling a $50\,\mathrm{kg}$ sled on flat snow. Your pulling force is $100\,\mathrm{N}$ at an angle of 37° above the horizontal. The kinetic friction coefficient is $\mu k=0.1$.
(a) Calculate the horizontal pull (F_x) and the vertical pull (F_y).
(b) Calculate the normal force from the snow.
(c) Find the net force in the X-direction.

Q5

A $50\,\mathrm{kg}$ skier is sliding down a ski slope tilted at 53°. The kinetic friction coefficient between the skis and the snow is $\mu k=0.5$.
(a) Calculate the force parallel to the slope and the force perpendicular to the slope respectively.
(b) What is the value of the normal force? Calculate the kinetic friction force.
(c) Calculate the acceleration of the skier sliding down the slope.

Q6

Imagine a 20kg lawnmower.

- Situation A:

You **pull** with a force of $100\,\mathrm{N}$ at an angle 37° *above* the horizontal.

- Situation B:

You **push** with a force of $100\,\mathrm{N}$ at an angle 37° *below* the horizontal.
(a) Draw the free-body-diagram (FBD) for both A and B situation. In which situation will the normal force be larger? Explain your reasoning based on the FBD.
(b) Which situation (Pushing or Pulling) will have a larger friction force pushing back against

you? Explain the reason.
(c) Calculate the exact normal force for Situation B.